Yi Xia *(Sean)*

Applied AI & ML Software Engineer

Open to: Applied AI · ML Engineering · Computer Vision · AI R&D · Data Science

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Applied AI engineer and PhD candidate who ships **production AI systems** across media and public-sector digital forensics. Four years at **TOHU MEDIA** delivering computer-vision pipelines, plus an AUT Research Assistant project for Tonga Police: an offline LLM system for analysing reports from the Autopsy digital-forensics platform and assessing risk. Hands-on across Python, PyTorch, Docker, local/cloud inference, end-user workflows and privacy-sensitive deployment. First author on all eight peer-reviewed papers, including two Q1 journal articles.

PRODUCTION AI
4 delivered AI systems across media and digital forensics.

ENGINEERING
Python, PyTorch, Docker, local LLMs and cloud GPUs.

RESEARCH DEPTH
8 first-author papers, 2 in Q1 journals; PhD thesis submitted.

MOBILITY
Based in Auckland; open to relocation.

CORE EXPERIENCE

Artificial Intelligence Developer
Feb 2022 - May 2026

TOHU MEDIA, Auckland - full-time June 2025 to May 2026; part-time alongside master's and PhD study before that

- **Owned the company's AI capability end-to-end**: dataset scope, PyTorch training/evaluation, Docker-packaged inference and deployment to company cloud GPU servers.
- Designed and shipped **three production AI systems** - object detection, OCR and action recognition - replacing manual transcription work on client video footage (specific deliverables under NDA).
- Engineered for messy real footage: variable cinematography, dense overlay text, low-contrast archives, long-form takes and workloads beyond single-workstation GPU limits.
- Worked directly with editors and the creative director on architecture, useful outputs and latency/accuracy trade-offs while preserving client data sovereignty.

Research Assistant - LLM Digital Forensics
Oct 2025 - Feb 2026

Auckland University of Technology · Client: Tonga Police

- Built and delivered **Autopsy AI**, a Windows application that parses Autopsy HTML case reports, combines rule-based risk scoring with LLM analysis, and generates investigator-ready reports.
- Designed a **100% offline local-LLM deployment** with Ollama/Gemma, Tonga-specific risk keywords and a standalone executable to keep sensitive forensic data on-device.

PhD Researcher - Computer Vision & Human Activity Recognition
Oct 2023 - present

Institute of Robotics & Vision, Auckland University of Technology

- Designed FATL-GCN for skeleton-based human activity recognition; published in **Pattern Recognition** (Q1, IF 7.6) in 2026.
- First-authored a generalisation survey in **Computer Science Review** (Q1 review, IF 17.5, 2026) and model papers at ICONIP 2024 and ICONIP 2026 (accepted); owned model design, GPU experiments, writing and rebuttal.

TECHNICAL CORE

Programming & ML
Python, Java, PyTorch, TensorFlow/Keras, scikit-learn, NumPy, pandas, Hugging Face

Computer Vision
YOLO family, object detection/tracking, OCR, video analytics, action recognition, segmentation, classification

Deployment
Docker, containerised inference, Linux, Git/GitHub, CUDA, cloud GPU servers, model evaluation

Research methods
GCNs, CNNs, attention/Transformers, skeleton-based HAR, frequency-domain learning

LLM Systems
Ollama/Gemma, Claude/OpenAI APIs, multi-provider integration, offline inference, prompt engineering

DOMAIN FIT

Computer Vision · **OCR / Document AI** · **Video Analytics** · **Digital Forensics** · **Public-Sector AI** · **Agritech** · **Applied Research**

ADDITIONAL EXPERIENCE

Research Assistant - Computer Vision
Nov 2022 - Jul 2023

School of Engineering, Computer & Mathematical Sciences, AUT

- Co-built KiwiDetector (improved YOLOv7) and KiwiTracker for real-time orchard video; lead paper has 50+ citations.
- Co-authored three peer-reviewed papers across IVCNZ 2022, SAI Intelligent Systems 2023 and PSIVT 2023.

Teaching Assistant
Mar 2025 - present

Auckland University of Technology

- TA on computer-science courses including programming, AI and computer vision; deliver classes and tutorials, prepare teaching materials and mark assessment.

Tech Service
Apr - Dec 2023

PB Tech, Auckland

- Handled hardware diagnostics, software troubleshooting, customer returns and custom PC builds; strengthened plain-English technical communication with non-specialist users.

Junior Lecturer / Co-Founder
2015 - 2021

Baiyin Vocational & Technology College / Baiyin Pinuo Training School, China

- Taught Computer Fundamentals, Java, IoT applications and RFID technologies to about 120 students per year for four academic years.
- Co-founded a private training school; built curriculum, ran academic operations and finance, then divested before relocating to New Zealand.

SELECTED PUBLICATIONS · 8 FIRST-AUTHOR PAPERS

Y. Xia, S. Yongchareon, R. Lutui, Q. Z. Sheng (2026). "Beyond Benchmark Accuracy: A Generalization-Centered Analysis of Deep Learning for Skeleton-Based Human Activity Recognition." *Computer Science Review*. Q1 review journal, IF = 17.5.
doi.org/10.1016/j.cosrev.2026.101030

Y. Xia, S. Yongchareon, R. Lutui, Q. Z. Sheng (2026). "Frequency-Aware Spatio-Temporal Topology Learning for Skeleton-Based Human Activity Recognition." *Pattern Recognition*. Q1 journal, IF 7.6.
doi.org/10.1016/j.patcog.2026.113146

Y. Xia, M. Nguyen, W. Q. Yan (2022). "A Real-Time Kiwifruit Detection Based on Improved YOLOv7." *IVCNZ 2022*, Springer LNCS. 50+ citations.
doi.org/10.1007/978-3-031-25825-1_4

Y. Xia, S. Yongchareon, R. Lutui (2026). "ESA-GCN: Efficient Semantic-Aware Graph Convolution for Skeleton-Based Action Recognition." *ICONIP 2026*, Springer LNCS, accepted, to appear.
shawn7702.github.io/esa-gcn/

Y. Xia, S. Yongchareon, R. Lutui, Q. Z. Sheng (2025). "Dynamic Self-Attention Gated Spatial-Temporal GCN for Skeleton-Based HAR." *ICONIP 2024*, Springer LNCS.
doi.org/10.1007/978-981-96-6588-4_14 · full list on Google Scholar

EDUCATION

PhD, Computer & Information Sciences
Auckland University of Technology

Oct 2023 - present. Thesis on deep learning for skeleton-based human activity recognition submitted; oral examination February 2027. AUT PhD Full Scholarship.

Master of Computer & Information Sciences
Auckland University of Technology

2021 - 2023. Specialisation: Computer Vision.

BEng, Logistics Engineering (Software)
Dalian Neusoft University of Information

2011 - 2015.

WHY THIS FITS

- **Role fit:** AI/ML engineering, software engineering for AI systems, computer vision, data-science engineering and applied research.
- **Location:** Auckland-based since 2021 with NZ experience across TOHU MEDIA, AUT and PB Tech; open to relocating.
- **Evidence of delivery:** four production AI systems, cloud GPU and offline LLM deployment, user-facing workflow integration.
- **Evidence of depth:** AUT PhD scholarship; eight first-author peer-reviewed publications, two in Q1 journals.

AWARDS & PROFESSIONAL SERVICE

- **AUT PhD Full Scholarship** - tuition + living stipend, Oct 2023 - Oct 2026.
- **SECMS Research Assistant Fund** and **AUT Summer Research Assistant Scholarship**.
- Peer reviewer for Pattern Recognition, ACM TOMM, Frontiers in Neurorobotics and ICONIP 2026.
- Member, AUT Institute of Robotics & Vision.

COMMUNICATION

- Tertiary teaching in China and New Zealand: four years as a lecturer in China, and seven AUT courses since 2025.
- Comfortable explaining AI systems to engineers, students, editors, managers and non-specialist stakeholders.
- Bilingual: Mandarin Chinese and professional English.

REFEREES

Dr Raymond Lutui, AUT, primary doctoral supervisor · raymond.lutui@aut.ac.nz
Associate Professor Sira Yongchareon, AUT, doctoral supervisor · sira.yongchareon@aut.ac.nz